F3111

Quantum and thermal Ising models

Jakub Matula Jan Müller Michal Zatloukal

AI: the new hope for scientific programming

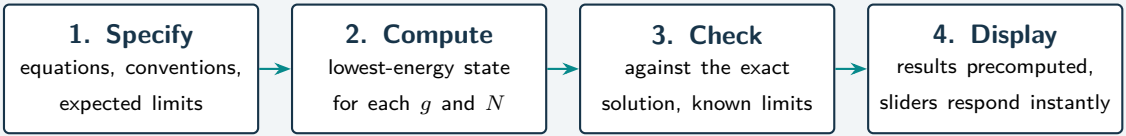
The Ising model and the quantities we measure

$$H(g) = -\underbrace{\sum_j \sigma_j^z \sigma_{j+1}^z}_{\text{neighbours prefer alignment}} - \underbrace{g \sum_j \sigma_j^x}_{\text{field favours the } x \text{ direction}}$$

- A quantum ring at zero temperature: small g favours z order; large g favours x polarization.
- The symmetric state has $\langle M_z \rangle = 0$ even when strongly ordered.
- We measure m_{rms} (order), m_x (field polarization), and G_{far} (distant correlations).

The infinite-chain critical field is $g_c = 1$; finite rings show a smooth crossover.

How we used the AI: specify, compute, check, display

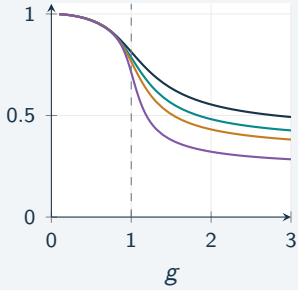


- **Tool:** GitHub Copilot, one stage at a time.
- **Why it is hard:** N spins give 2^N configurations; $N = 18$ gives $262\,144$.
- **What helps:** the energy table is almost all zeros, and symmetry halves it.

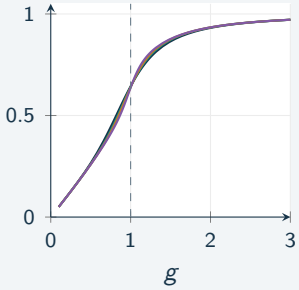
One stage at a time, each checked before the next.

Results: order, polarization and response

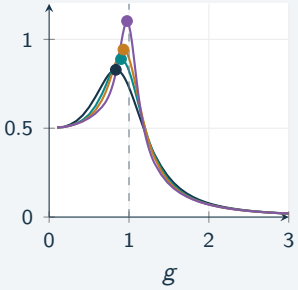
$m_{\mathrm{rms}} = \sqrt{\langle M_z^2 \rangle}$



m_x



$\chi_x = dm_x/dg$

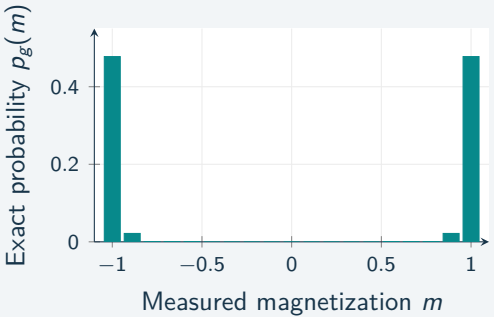


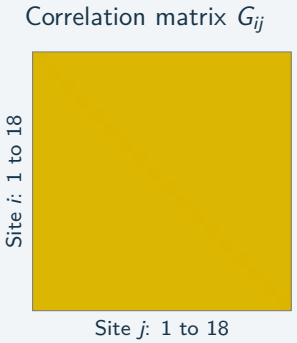
$N = 6$ $N = 8$ $N = 10$ $N = 18$ Dashed: $g_c = 1$

The crossover sharpens with N. Dots mark refined response maxima.
For a finite chain, the strong-field limit is $m_{\mathrm{rms}} \to 1/\sqrt{N}$.

Weak field: strong order without choosing a sign

$N = 18$, $g = 0.2$ $\quad m_{\mathrm{rms}} = 0.995$ $\quad m_x = 0.101$ $\quad G_{\mathrm{far}} = 0.990$



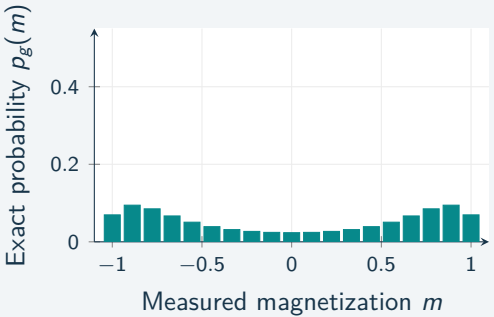


Fixed scale: black $= -1$, white $= 0$, yellow $= 1$

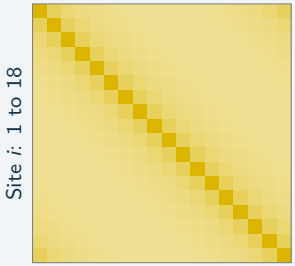
Probability is concentrated near all-up and all-down outcomes.
Spins remain correlated even across the ring.

Near the transition: a broad distribution

$N = 18$, $g = 1.0$ $\quad m_{\mathrm{rms}} = 0.712$ $\quad m_x = 0.637$ $\quad G_{\mathrm{far}} = 0.417$



Correlation matrix G_{ij}

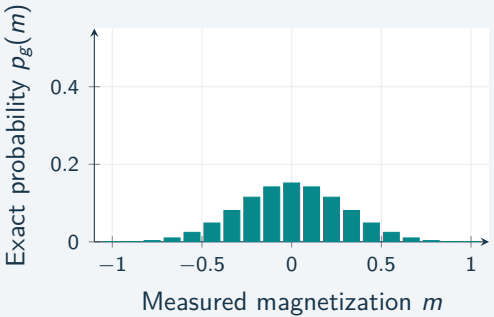


Fixed scale: black $= -1$, white $= 0$, yellow $= 1$

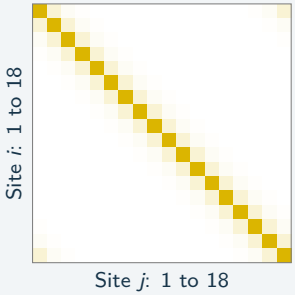
Intermediate magnetizations gain weight, while long-distance correlations weaken.
The finite ring still retains substantial order.

Strong field: mixed z outcomes, large x polarization

$N = 18$, $g = 3.0$ $\quad m_{\mathrm{rms}} = 0.284$ $\quad m_x = 0.972$ $\quad G_{\mathrm{far}} = 1.93 \times 10^{-5}$



Correlation matrix G_{ij}



Fixed scale: black $= -1$, white $= 0$, yellow $= 1$

The distribution is concentrated nearer zero and distant correlations are tiny.
Mixed z measurements coexist with strong polarization along x.

Results: locating the crossover

The response $\chi_x = dm_x/dg$ is largest where the polarization changes most rapidly.

N	Coarse g^*	Refined g^*	Change	Peak χ_x
6	0.840	0.835	-0.005	0.829
8	0.900	0.900	0.000	0.889
10	0.920	0.930	$+0.010$	0.943
18	0.980	0.975	-0.005	1.103

The peaks grow and move toward $g_c = 1$. The local refined spacing is 0.005.

Interpretation: $g^*(N)$ is a finite-size indicator. These four sizes do not establish an extrapolated critical point.

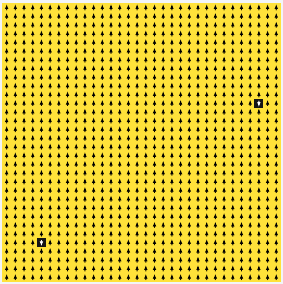
Thermal disorder and ordering on cooling

$T = 4.0,\ |m| = 0.014$



Disordered

$T = 1.5,\ |m| = 0.996$



Ordered after cooling

The 32×32 classical model orders through local Metropolis updates.
Infinite-lattice critical temperature: $T_c \approx 2.269$; this is separate from quantum g.

Arrows show local spins: yellow points up, black points down.

Strengths and limitations of our approach

Strengths

- Detailed prompts keep the implementation tied to the physics.
- Independent checks make numerical errors easier to detect.
- Precomputation makes the quantum display responsive.

Limitations

- AI-generated code still needs expert review and validation.
- The quantum state space grows exponentially with N.
- Small finite systems only indicate the transition region.

Numerical checks: sparse/dense energy difference $\approx 8 \times 10^{-15}$; reported normalized eigenpair residuals $< 2.8 \times 10^{-11}$.